Department of Information Technology

Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G.Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Directed Acyclic Graph
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO16
- **Activity Chosen:** One Minute Paper

- **Justification:**

  Directed Acyclic Graph is an important topic comes under the category of Graph Data Structure. A directed acyclic graph (DAG) is a conceptual representation of a series of activities. The order of the activities is depicted by a graph, which is visually presented as a set of circles, each representing an activity, some of which are connected by lines, representing the flow from one activity to another. Directed acyclic Graph have direct applications in the design of networks, including computer networks, telecommunications networks, Route planning and navigation and Web page ranking.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

  At the end of the class, the students were asked to write about the Directed Acyclic Graph topic discussed in the class. During the first 40 minutes instructor explained the particular concepts/topic in classroom. The students expressed the understood concepts in the minimum spanning tree topic and the content they need more clarification in that particular topic. The points on Directed Acyclic Graph topic shared by the students through this activity as shown in the figure 1. This activity shows whether the students can able to understand the concept of Directed Acyclic Graph and the steps and procedure of topological ordering to find the order to visit the vertex, this activity also shown their involvement in the particular class.

- **CO – PO / PSO mapping:**

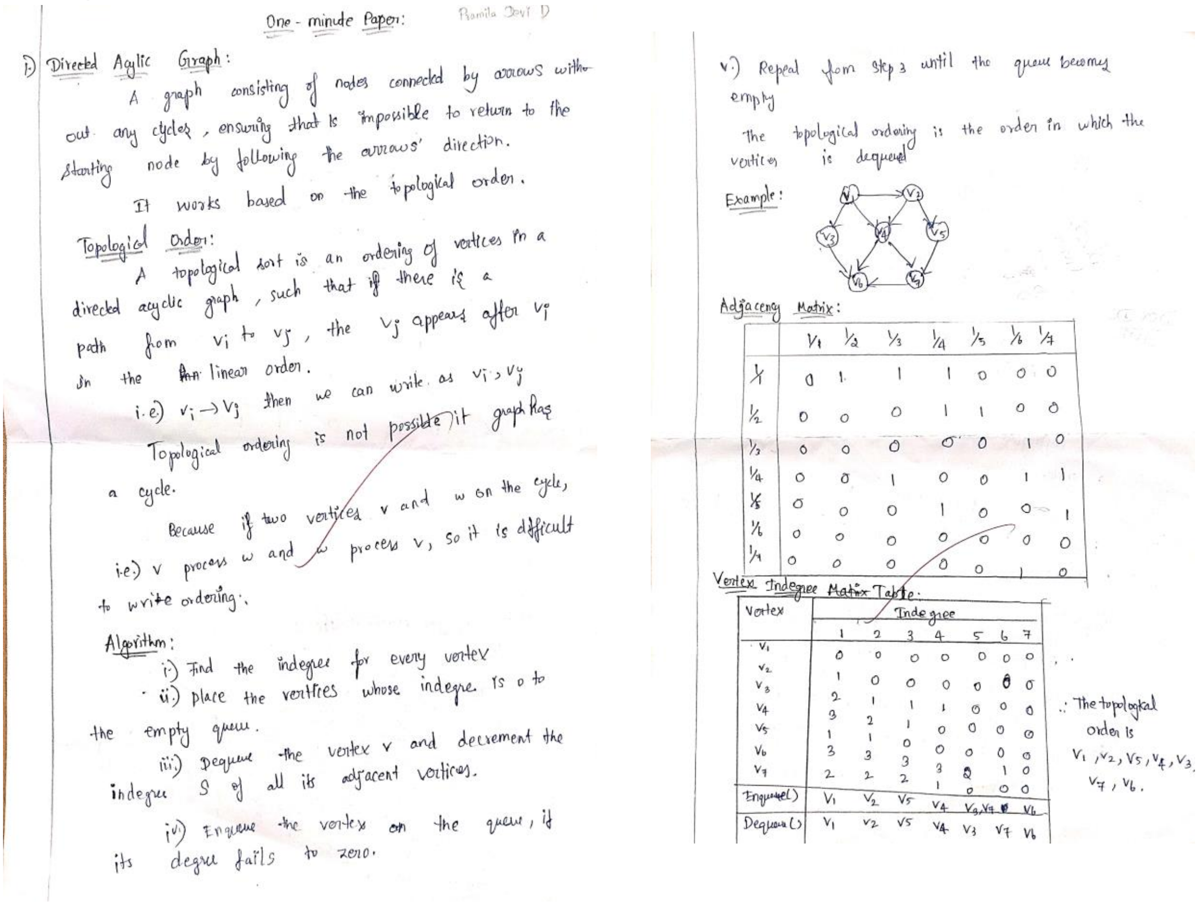
CO	PO1	PO2	PO3	PO9	PSO1
CO5	3	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	1	1	1	1
Justification for correlation	Needs engineering and mathematical knowledge for understand DAG concept	Various algorithms used in analysis of solving complex engineering problems	The students can able to find solution for the complex nonlinear problems using the concept of DAG.	The students individually involved in this activity	The students can apply the concept of DAG in solving various problems of data science and algorithms.

- **Images / Screenshot of the practice:**


One - minute Paper: Pramila Devi D

i) Directed Acylic Graph:
A graph consisting of nodes connected by arrows without any cycles, ensuring that is impossible to return to the starting node by following the arrows' direction.
It works based on the topological order.

Topological Order:
A topological sort is an ordering of vertices in a directed acyclic graph, such that if there is a path from v_i to v_j, the v_j appears after v_i in the linear order.
i.e) $v_i \rightarrow v_j$ then we can write as $v_i > v_j$
Topological ordering is not possible if graph has a cycle.
Because if two vertices v and w on the cycle, i.e) v process w and w process v, so it is difficult to write ordering.

Algorithm:
i) Find the indegree for every vertex
ii) place the vertices whose indegree is 0 to the empty queue.
iii) Dequeue the vertex v and decrement the indegree's of all its adjacent vertices.
iv) Enqueue the vertex on the queue, if its degree falls to zero.
v) Repeat from step 3 until the queue becomes empty
The topological ordering is the order in which the vertices is dequeued

Example:

Adjacency Matrix:

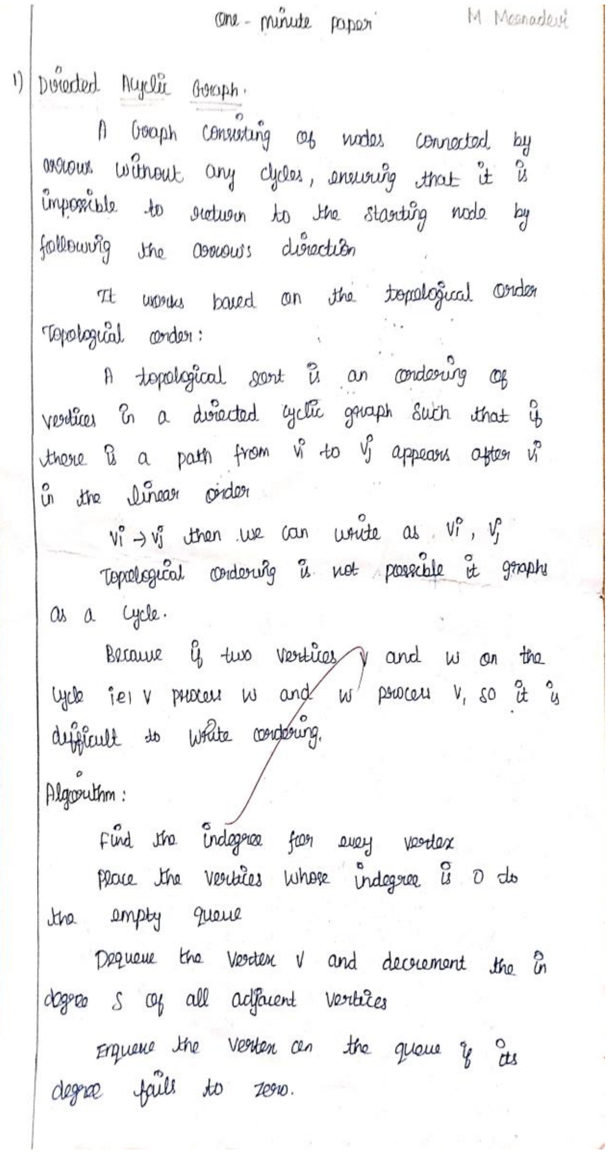
	V1	V2	V3	V4	V5	V6	V7
V1	0	1	1	1	0	0	0
V2	0	0	0	1	1	0	0
V3	0	0	0	0	0	1	0
V4	0	0	1	0	0	1	1
V5	0	0	0	1	0	0	1
V6	0	0	0	0	0	0	0
V7	0	0	0	0	0	1	0

Vertex Indegree Table:

Vertex	Indegree 1	2	3	4	5	6	7
V1	0	0	0	0	0	0	0
V2	1	0	0	0	0	0	0
V3	2	1	1	1	0	0	0
V4	3	2	1	0	0	0	0
V5	1	1	0	0	0	0	0
V6	3	3	3	3	2	1	0
V7	2	2	2	1	0	0	0
Enqueue( )	V1	V2	V5	V4	V3, V7		V6
Dequeue( )	V1	V2	V5	V4	V3	V7	V6

∴ The topological order is V1, V2, V5, V4, V3, V7, V6.

Figure 1. a Sample One-Minute Paper of Pramila Devi. D



One - minute paper — M Meenadevi

1) Directed Acyclic Graph.

A Graph consisting of nodes connected by arrows without any cycles, ensuring that it is impossible to return to the starting node by following the arrows direction

It works based on the topological order

Topological order:

A topological sort is an ordering of vertices in a directed cyclic graph such that if there is a path from vi to vj appears after vi in the linear order

vi → vj then we can write as vi, vj

Topological ordering is not possible it graphs as a cycle.

Because if two vertices v and w on the cycle ie, v proces w and w proces v, so it is difficult to write ordering.

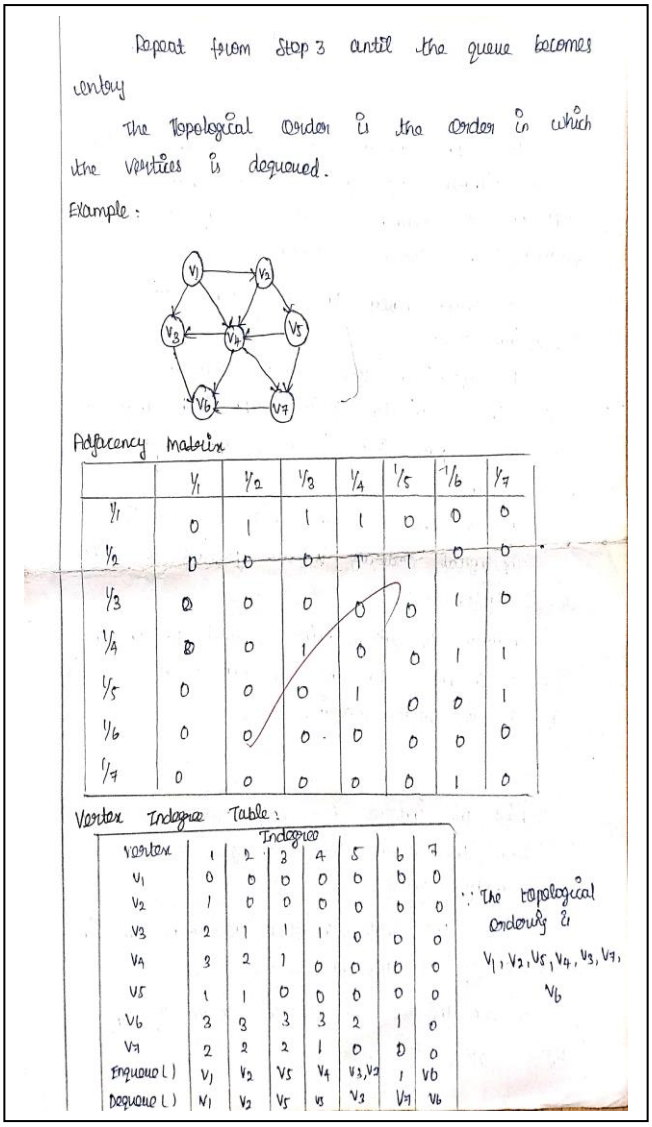
Algorithm:

Find the indegree for every vertex

Place the vertices whose indegree is 0 to the empty queue

Dequeue the Vertex V and decrement the in degree s of all adjacent vertices

Enqueue the vertex on the queue if its degree fails to zero.



Repeat from Step 3 until the queue becomes empty

The Topological Order is the order in which the vertices is dequeued.

Example:

Adjacency matrix

	V1	V2	V3	V4	V5	V6	V7
V1	0	1	1	1	0	0	0
V2	0	0	0	1	1	0	0
V3	0	0	0	0	0	1	0
V4	0	0	1	0	0	1	1
V5	0	0	0	1	0	0	1
V6	0	0	0	0	0	0	0
V7	0	0	0	0	0	1	0

Vertex Indegree Table:

	Indegree						
vertex	1	2	3	4	5	6	7
V1	0	0	0	0	0	0	0
V2	1	0	0	0	0	0	0
V3	2	1	1	1	0	0	0
V4	3	2	1	0	0	0	0
V5	1	1	0	0	0	0	0
V6	3	3	3	3	2	1	0
V7	2	2	2	1	0	0	0
Enqueue ( )	V1	V2	V5	V4	V3, V7		V6
Dequeue ( )	V1	V2	V5	V4	V3	V7	V6

∴ The topological ordering is V1, V2, V5, V4, V3, V7, V6

Figure 1.b Sample One-Minute Paper of Meenadevi. M

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - The students felt that they can able to understand the events topic clearly after conducting this activity.
- ***Benefit of the practice:***
  - Students were actively participated in this activity.
  - From this activity, the students can get more clarity in the particular topic.
  - Students can able to explain the topic without any confusion.

- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.

- ***Challenges faced in implementation:***
  - Some students are hesitated and lack to write the concepts that they learnt in the class.

References:

- https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html
- https://www.unl.edu/gradstudies/current/teaching/minute
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1 GM CS335 3_ Functions_OMP.pdf
- https://oncourseworkshop.com/self-awareness/one-minute-paper/

Signature of Faculty Member **HOD**